

CINDER Closure-Conditioning and Multiroot Study Summary

Purpose

This report packages the main findings from the closure-conditioning study and the follow-on multiroot exploration so the project can be picked back up quickly later. The focus was the fully engaged contact closure: conditioning of the 8x8 mechanics, uniqueness of accepted roots on representative states, and the existence and interpretation of global multiroot structure in both stick-stick and one-contact closures.

Headline findings

- Multistart robustness was strong: 15 representative/shared states were tested with 375 total starting points; all accepted solutions within each state collapsed to exactly one distinct physical root.
- Across the practical physical and expanded maps, the 8x8 mechanical system stayed solvable everywhere sampled; no rank-deficient grid points appeared outside the deliberately far-out broad domain.
- Stick-stick multiroot structure exists. The best visual example occurred at `sf0p950_sdotm40_rpm3000` with fold prominence 7.47 Nm and root separation 0.220.
- One-contact multiroot structure also exists, but was much rarer. Across 62 frozen states and all mixed-branch searches, only one unique geometry was found; its two-root slice sat only about 1.02 Nm from the fold and one root lay very near lift-off.
- Practical solver conclusion: follow the local branch continuously, detect approach to folds via small slope / large force discontinuity markers, and treat branch switching as a boundary or regime-transition issue rather than a generic operating mode.

Study design and what was tested

Closure-conditioning core

- Representative frozen states were assembled from shared/default cases: seated low-ratio, mid-shift, upper stop, and free-shift opening, alongside shared operating cases imported from the global case library.
- For each state, λ_p - λ_s maps were built over physical, expanded, and sometimes broad domains. The maps recorded residuals, 8x8 conditioning, stick Jacobian conditioning, inadmissibility, and topology/mechanism failure regions.
- Multistart solves then tested whether multiple accepted physical roots were actually reached from different initial guesses on each frozen state.

Follow-on multiroot exploration

- When the maps suggested folded zero contours, continuation scripts were used to follow branches and build more interpretable visuals: λ -plane overlays, one-dimensional crossing slices, branch-continuation plots, and 3D manifold views.
- Two separate questions were explored: (1) stick-stick multiplicity, where the full closure is satisfied by two different trial utilization pairs, and (2) true one-contact multiplicity, where one interface is prescribed sliding and the remaining scalar sticking residual itself develops a fold.

What to keep in mind

The maps are global diagnostic tools, not predictions that the time simulation actually visits every plotted point. They are most useful for understanding what kinds of numerical or physical pathologies are possible, where they live, and whether the actual simulation trajectory is likely to encounter them.

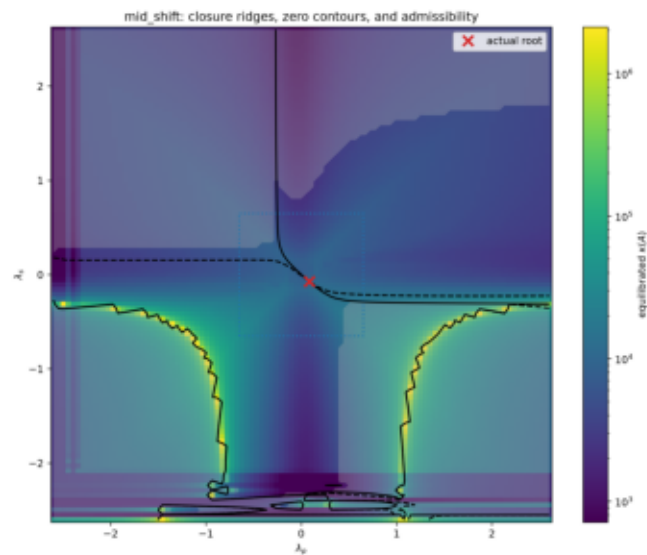
Representative physical-domain conditioning summary

State	Topo. admissible frac.	Full static frac.	Max cond(A)	Max cond(J)
low_ratio_seat	0.424	0.424	1.33e+04	1.01e+04
mid_shift	0.879	0.879	3.7e+04	2.66e+05
upper_stop	0.616	0.616	3.38e+05	1.34e+05
free_shift_opening_70	0.857	0.857	8.26e+03	6.97e+04

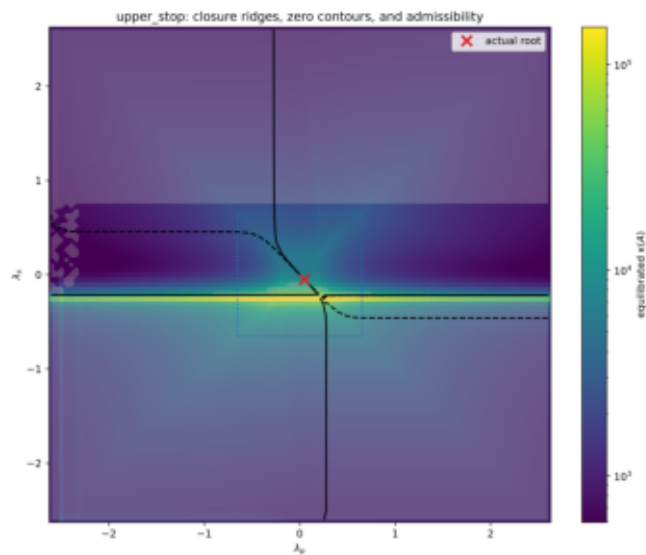
Interpretation: the practical physical domains were largely admissible, and the largest scaled 8x8 condition number among these representative physical maps was about 3.4e+05. The more dramatic structures emerged when the search was intentionally expanded into much larger λ space, especially near inadmissible or topology-violating regions.

Representative λ -space structure

Mid-shift expanded feature overlay



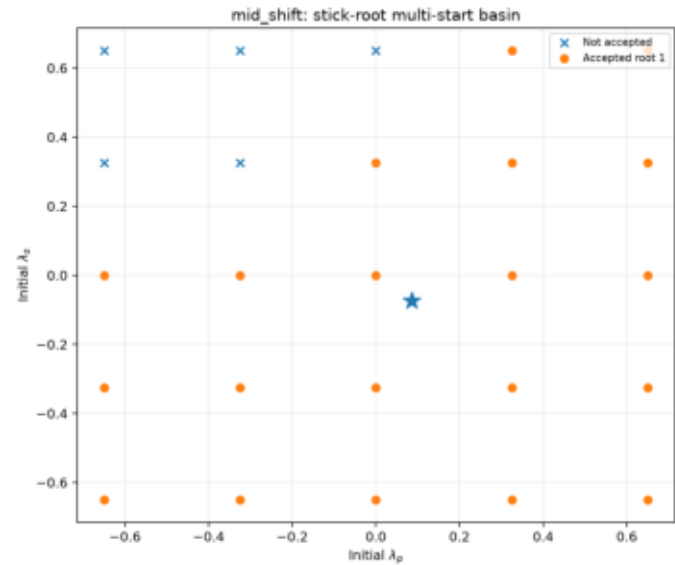
Upper-stop expanded feature overlay



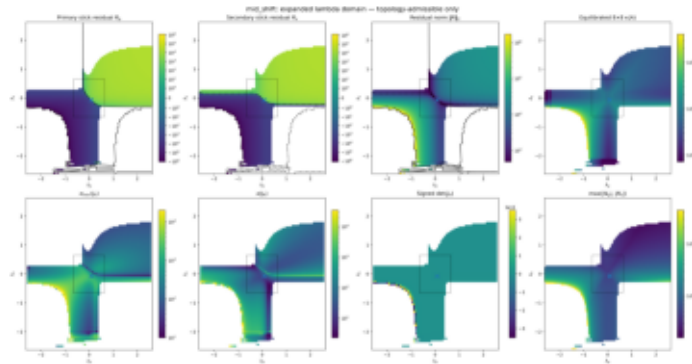
These overlays are useful because they show where the interesting zero-contour and conditioning structures actually live. The main takeaway from the study was that the visually dramatic ridges and asymptotes are not random: they line up with places where the 8x8 response becomes very sensitive, or where a trial point is pushing against physical/topology limits. Those structures are therefore informative, but they are not automatically operationally relevant.

Accepted-root uniqueness on representative states

Mid-shift multistart basin

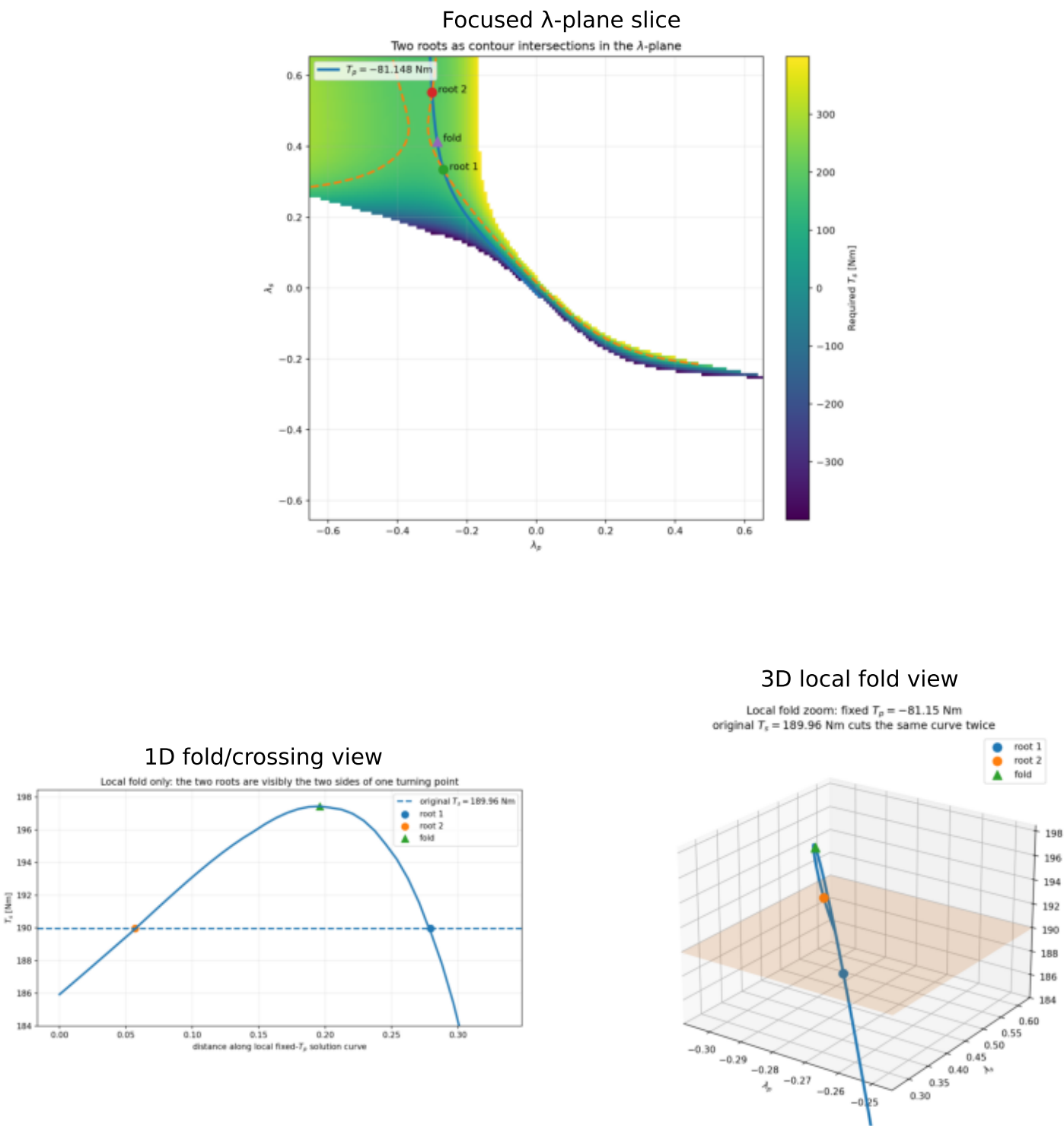


Mid-shift conditioning map (topology masked)



Multistart testing was reassuring. Across 15 representative/shared states, all accepted roots within each state collapsed to one distinct physical solution. This does not prove global uniqueness everywhere, but it does show that the solver was not secretly bouncing among several accepted roots on the practical test cases.

Stick-stick multiroot: best visual example



The best stick-stick visual example occurred at sf0p950_sdotm40_rpm3000. It showed a clear folded geometry: a continuous solution surface bent back on itself, so a fixed torque slice intersected it twice. Quantitatively, the chosen example had fold prominence 7.47 Nm and root separation 0.220. The minimum static margin along the selected two-root example was 0.098, so the example was physically admissible but still boundary-adjacent enough to explain why the global maps only revealed such multiplicity in selective regions.

How to interpret the stick-stick manifold result

Plain-language interpretation

The important idea is that the two plotted roots are not best thought of as two unrelated answers. They are two intersections with the same folded solution sheet. If a torque slice cuts the folded sheet once, there is one root; if it becomes tangent, a double root is born; and if it cuts through both arms, there are two roots. That is why the 3D view was useful: it makes clear that the two roots belong to one bent manifold rather than two disconnected phenomena.

Numerical implication

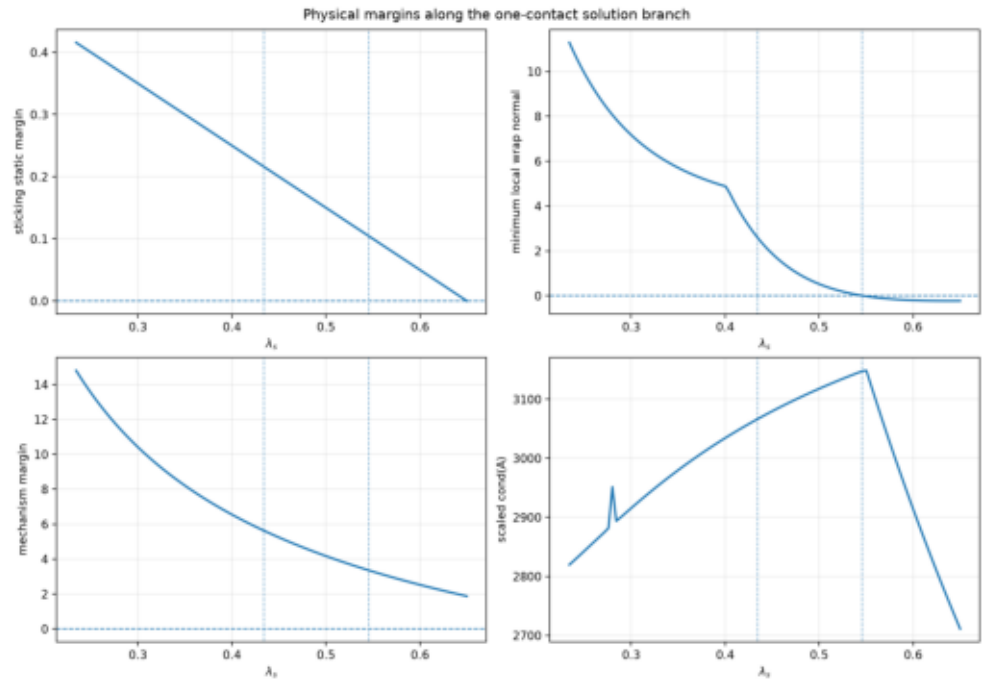
- A time simulation that follows the existing stick-stick solution continuously should normally remain on the same branch.
- Potential ambiguity appears mainly when entering stick-stick from another regime, or when the followed branch approaches inadmissibility / a failure boundary.
- If a branch disappears at a fold, the state can remain continuous while the supporting contact forces jump to the other admissible branch or trigger a regime transition. That is a force-level discontinuity story, not necessarily a state-jump story.

Practical conclusion

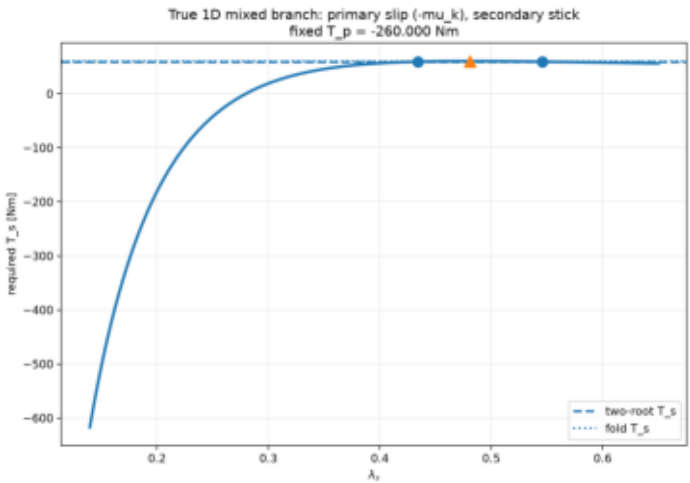
This is enough to justify continuation-minded logic in the solver and to motivate branch/fold diagnostics. It is not a reason to believe the nominal simulation is fundamentally ambiguous most of the time.

True one-contact multiroot: scalar fold example

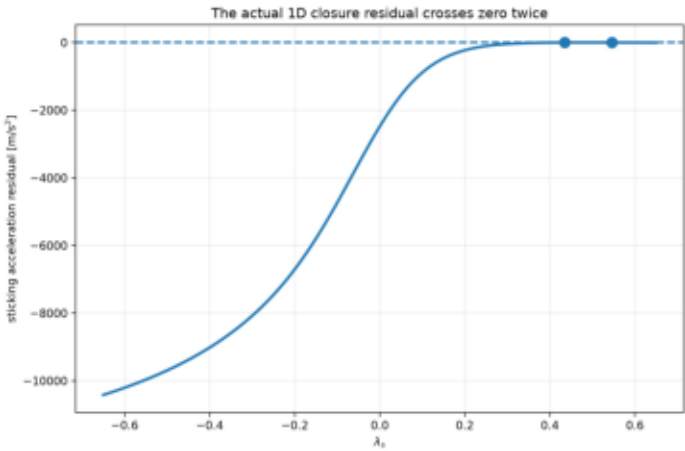
Branch physics along the fold search



Required torque fold



Scalar residual with two roots



The one-contact search found a genuine scalar-fold example on branch "primary slip ($-\mu_k$), secondary stick" at state sf0p950_sdotm40_rpm2500. The varied torque slice showing two roots sat only 1.02 Nm away from the fold. The two roots were $\lambda = 0.434$ and 0.546 , with residual slopes 54.6 and -28.0. The fold itself was healthy, but the second root approached lift-off: its minimum local wrap normal was only 0.0062.

How to interpret the one-contact result

What was actually found

- The search scanned 62 frozen states and found 15 candidate rows, but these reduced to only 1 unique geometry(ies). In practice, the table repeated the same fold while varying a fixed torque that did not materially change the branch geometry.
- This means one-contact multiplicity is real, but much rarer than a raw candidate count might suggest.

Why the GIF looked delicate

The 1D scalar residual really did have two roots, but the displayed two-root slice lived very close to the saddle-node creation point, and one root was near local contact loss. So visually it looks almost parallel to the zero line: not because the result is fake, but because the useful positive lobe is small compared with the full residual scale.

Practical solver take

- This supports the same continuation policy as the stick-stick case: do not use a global root picker that can hop across disconnected parts of λ space.
- Treat one-contact multiplicity as a rare edge-case diagnostic rather than an everyday operating hazard.

Overall conclusions and pickup-later checklist

Overall conclusions

- The closure-conditioning study achieved its main goal: it separated “interesting pictures in trial λ space” from “things the actual solver is likely to suffer from.”
- Most alarming conditioning structures concentrate near inadmissible or topology-limited regions rather than ordinary trajectories.
- Representative multistart tests were strongly single-root within the accepted physical set.
- Global multiplicity does exist in both stick-stick and one-contact closures, but the examples found were rare, structured, and typically boundary-adjacent. They support careful branch-following logic rather than suggesting pervasive nonuniqueness.
- A useful diagnostic distinction emerged: large jumps may appear more naturally in contact forces / reaction quantities than in the core state itself. That is both physically meaningful and numerically useful for detection.

Suggested next actions when revisiting the study

- If needed, add an in-simulation discontinuity monitor on closure forces / reactions as a practical fold-boundary detector.
- Keep the global/default case library shared between invariants and closure-conditioning so new studies reuse the same representative states.
- Use the stick-stick and one-contact fold story artifacts as visual references when writing the final results section.
- If more work is needed, the next high-value extension would be a short writeup that translates these findings directly into solver policy: initialization, continuation, event handling, and boundary-transition logic.

Key artifact roots

/mnt/data/artifacts(10).zip

/mnt/data/best-visual-multiroot-story(1).zip

/mnt/data/one-contact-multiroot-fold-search.zip

/mnt/data/best_story_final_zoom